

GRAFT: Symmetry-Aware Atom Mapping through Continuous Fragment Growth

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Atom-to-atom mapping for reaction analysis requires retaining alternative correspondences without enumerating redundant atom permutations. We introduce GRAFT, a graph-based method that grows matched fragments on weighted reactant and product graphs, represents compatible placements through correlated symmetry actions, and branches both on saturated placements and on competition between fragments. Completed searches are deduplicated by unordered matched fragment pairs while preserving their internal mapping alternatives. A symbolic decoder extracts distinct signed bond-event patterns directly from these families, separating the tolerance used for matching from the thresholds used to interpret bond changes. On all 1,851 Golden benchmark records, bidirectional search with edge sweeps recovers an annotated correspondence among its retained alternatives in 1,834 cases (99.08%) with one seed ordering per cut and 1,840 (99.41%) with ten; two ten-order cases remain unresolved. On 140 coordinate/Wiberg-bond-order reaction pairs, the configuration with fragment competition recovers 166 of 168 minimum-event classes among the returned SLAP-sweep witnesses, with complete reference-class coverage in 139 reactions. Final fragment deduplication reduces 237,645 ordered branch records to 124,641 unordered combinations. GRAFT provides an inspectable representation of mapping and bond-event alternatives for subsequent chemical assessment.

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1 Introduction

Atom-to-atom mapping assigns corresponding atoms across a reaction. It underlies reaction-center identification, template extraction, and the preparation of endpoint structures for mechanistic calculations. A reaction can admit several plausible correspondences, while many different atom-index assignments are chem-

ically equivalent. An algorithm intended to retain alternatives must therefore resolve two questions: which structural choices deserve separate candidates, and which permutations can be represented together?

Existing approaches use learned correspondence signals or graph-based optimization. RXNMapper derives mappings from attention in a reaction language model (9); LocalMapper learns from chemist-labeled reactions (1); and SLAPMapper combines iterative label refinement with sequential linear assignment (5). The Golden benchmark provides a common set of annotated reactions for assessing mapping recovery (6). For applications requiring alternatives, recovering the annotation somewhere in a returned family is a different objective from producing one correct top-ranked mapping.

Weighted endpoint graphs introduce a further distinction. A permissive matching tolerance can preserve a fragment despite changes in its bond orders, but the corresponding atom shuffles need not preserve the bond events assigned at a finer threshold. Replacing a compressed family by one representative can therefore hide distinct transformations. Likewise, greedily growing one fragment can prevent a neighboring fragment from reaching an alternative assignment.

GRAFT addresses these issues by branching over structural fragment choices while keeping symmetry-related atom bijections compressed (Figure 1). Continuous growth, alternative placements, and fragment priorities determine the structural choices. Its final representation groups unordered matched fragment pairs and retains the allowed mapping alternatives within each group. A constraint-based decoder projects these families onto distinct bond-event patterns. The method searches for structurally compatible alternatives without requiring the reference transformation to minimize the number of bond changes.

2 Method

2.1 Weighted matching and continuous growth

Let V_R, V_P be the reactant and product atom sets, Z_R, Z_P their element labels, and W_R, W_P their symmetric bond-weight matrices. A partial mapping $m : D \subseteq V_R \rightarrow V_P$ is injective and element compatible. On balanced endpoints, a completed mapping is a bijection. Explicit hydrogen atoms participate in the search. Molecular coordinates can accompany the graphs but do not determine the growth decisions.

The active source graph contains edges with weight at least b . Starting from a seed atom, GRAFT grows a connected source fragment F while carrying multiple compatible product placements. A frontier queue prioritizes stronger source edges. To add source atom u with product image v , a placement must satisfy

$$Z_R(u) = Z_P(v), \quad v \notin m(F), \quad W_P(v, m(r)) \geq b, \quad |W_R(u, r) - W_P(v, m(r))| \leq \tau_{\text{iso}} \quad (1)$$

for every active, nonrelaxed source edge (u, r) into F . All relevant fragment bonds are checked, rather than only the frontier edge that proposed the extension. Source nonedges do not impose negative matching constraints, so additional product bonds remain possible.

An accepted extension enlarges the fragment and filters its compatible placements. Growth continues even if only one placement remains. An unsupported edge is deferred, and other frontier edges can still be explored. When growth reaches an established fragment, ordinary continuation respects its existing assignments. A fragment saturates when no further admissible extension remains; structurally distinct surviving placements then generate separate continuations for the remaining atoms.

This is a greedy search along each trajectory: an accepted extension can remove placements that would support a different fragment boundary. GRAFT broadens coverage through seed orderings, single-edge constraint relaxations, and local changes in fragment priority. A sweep combines the uncut source graph with searches in which one eligible source edge is omitted. Bond-event evaluation always uses the original endpoint matrices.

2.2 Automorphisms and correlated mapping families

The distinction between structural branches and explicit bijections is central to the design. Even one fragment family can contain combinatorially many assignments: k independently exchangeable methyl-hydrogen

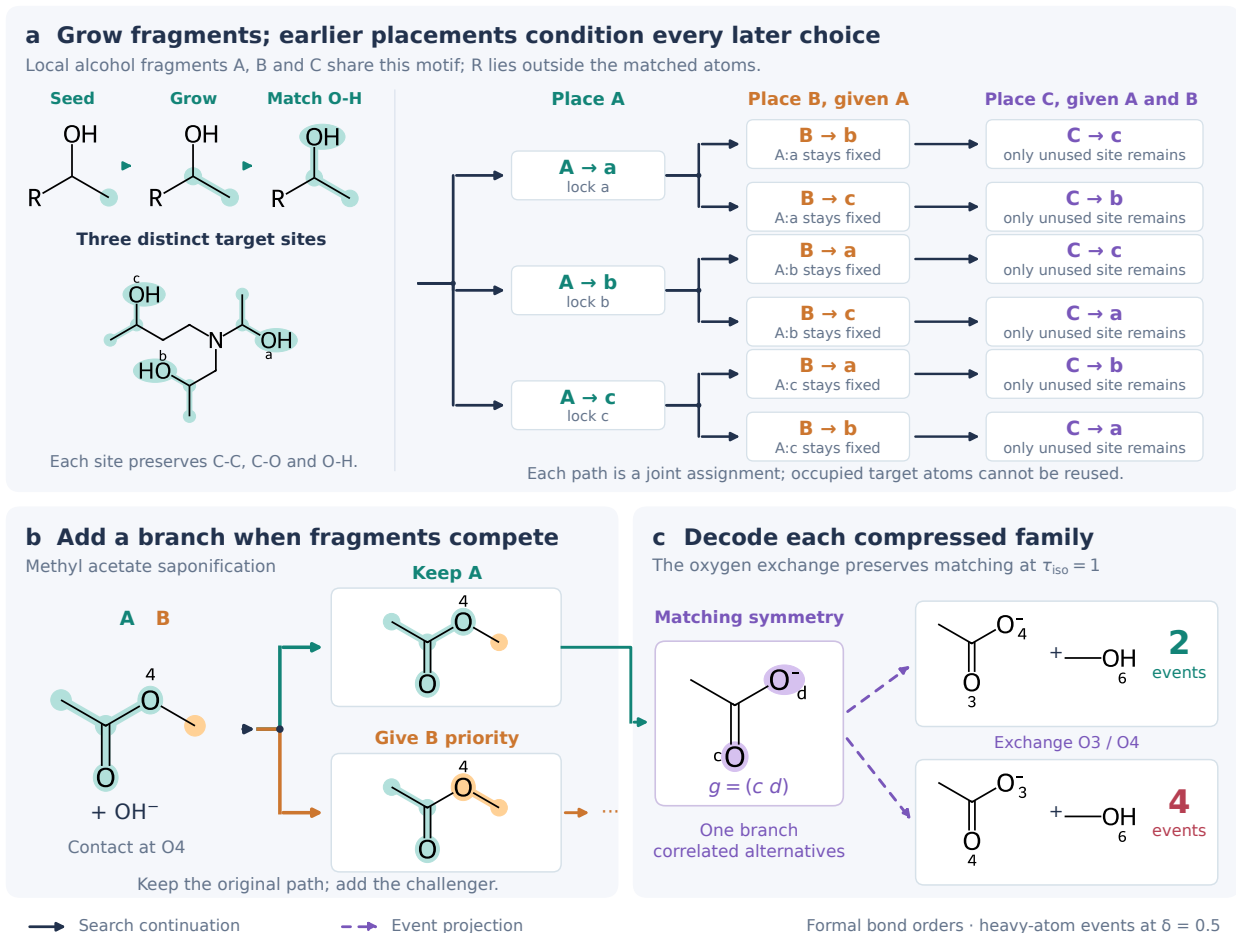


Figure 1 Conditional fragment branching and compressed symmetry. (a) A local alcohol motif grows while retaining its O-H bond. Three sites in the depicted target have distinct molecular environments. For three source fragments sharing this motif, placing A fixes one site; B then branches over the two unused sites, and C takes the remaining site. The tree illustrates local conditioning and injectivity; R denotes atoms outside the matched subgraph, and this is not a complete reaction-search trajectory. All attached hydrogens in the illustrated local matches are checked. (b) At a fragment contact in methyl acetate saponification, the original path retains O4 in A. An added path gives B priority, rematches the contact, and regrows displaced assignments. (c) Exchanging the two acetate oxygen images preserves the fragment’s element and bond-weight matching tests at $\tau_{iso} = 1$, but changes two heavy-atom events into four at threshold 0.5. Neither acetate oxygen carries H. Dashed arrows are event projections within a family, not additional search branches. Saved constraints govern allowed symmetry actions; event counts here show the heavy-atom projection.

triplets admit $(3!)^k$ bijections. Expanding those automorphisms into separate mappings would multiply the work without adding fragment choices. GRAFT instead stores the permitted permutation actions and their constraints, and decodes event alternatives directly from this representation. Branch caps bound retained alternatives within each growth call; they do not guarantee a small global search. This avoids explicitly listing symmetry-related bijections within each retained family.

Compatible placements are stored using a representative assignment, unresolved assignment blocks, and permutation generators. During growth, graph canonicalization accounts for the current fragment, locked assignments, and boundary information. Each saved transition records its action generators together with conditioning and fragment-preservation constraints; their conjunction defines the allowed mapping relation. Canonical labeling and automorphism computation use nauty (7).

Matching tolerance and automorphism equivalence have different roles. The relation $|x - y| \leq \tau_{\text{iso}}$ is generally not transitive, so it does not by itself define a permutation group. To describe an intrinsic symmetry shared by a final fragment pair, we instead color product bonds by their responses to the fragment’s matching tests. For a product weight w and the set $\mathcal{W}$ of active reactant weights for the same element pair, the label records

$$\chi_{\mathcal{W}}(w) = (\mathbf{1}[w \geq b], (\mathbf{1}[w \geq b \wedge |w - v| \leq \tau_{\text{iso}}])_{v \in \mathcal{W}}). \quad (2)$$

An exact automorphism of this colored graph preserves every encoded matching test. This definition depends on the final fragment pair, not its growth order.

The saved mapping relation also contains the constraints imposed by its search decisions. We represent a flat family $\mathcal{F}_j$ by a seed mapping m_j , a sequence of allowed action groups $G_{j,1}, \dots, G_{j,k}$, and final preservation constraints C_j :

$$\mathcal{F}_j = \{a_k \circ \dots \circ a_1 \circ m_j : a_i \in G_{j,i}, C_j(a_k \circ \dots \circ a_1 \circ m_j)\}. \quad (3)$$

These actions encode correlated moves; atoms in a common orbit are not independently permutable. Disjoint actions commute and can be placed in a canonical order. The order of overlapping actions is retained. A sequence of groups is not generally equivalent to the group generated by their union.

The intrinsic automorphism of a fragment pair is shared metadata, not permission to enlarge its saved family. Multiple admissible seed mappings can occupy different intrinsic symmetry orbits, and different search constraints can restrict the allowed actions. GRAFT therefore retains these alternatives explicitly inside the compressed representation.

2.3 Branching through fragment competition

Saturated-placement branching explores different images of a fragment. Fragment competition additionally explores a different division of the source atoms. Suppose established fragment A borders fragment B . In a challenger branch, GRAFT rematches B together with a contacting atom of A , allowing the challenger to occupy a different compatible product region. Assignments conflicting with the new placement are released. Unaffected assignments remain anchored, and the fragment-growth procedure fills the resulting gaps. Small attached components can also be released when their assignments depend on the displaced region.

The repaired continuation begins from the revised assignments and constraints; it does not inherit displaced assignments as immutable conditions. Before admission, its complete representatives are checked for element compatibility, injectivity, and preservation of their required fragment bonds. Accepted continuations join the original output, so competition supplements the available alternatives. A finite queue, a bound on successive priority changes, and a completion budget control its cost. Repeated proposals and equivalent completion requests are cached.

2.4 Unordered final branches

During search, history and boundary conditions matter because they determine valid continuation. At completion, the branch identity is instead the unordered collection of matched atom sets,

$$\mathcal{B} = \{(R_1, P_1), \dots, (R_q, P_q)\}, \quad P_i = m(R_i), \quad (4)$$

under a fixed matching policy. Sorting atoms within each set and then sorting the fragment pairs removes dependence on growth order and fragment labels. Growing A before B or B before A yields one branch when the final paired sets agree.

Each branch stores the union of its flat families. Identical normalized family records are shared, while distinct constraints or action programs remain available. This transformation preserves the represented mapping union: it changes its grouping, not its admissible assignments. Provenance links retain the discovering paths, but the original search hierarchy is unnecessary for decoding. This conservative deduplication does not assert that the remaining families are mutually disjoint.

2.5 Decoding distinct bond-event patterns

For each complete mapping, define the signed change for an unordered source pair $\{u, v\}$ by

$$\Delta_{uv}(m) = W_P(m(u), m(v)) - W_R(u, v), \quad s_{uv}(m) = \begin{cases} +1, & \Delta_{uv}(m) \geq \delta_{uv}, \\ -1, & \Delta_{uv}(m) \leq -\delta_{uv}, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Positive events denote bond formation or strengthening; negative events denote cleavage or weakening. The event count is $E(m) = \sum_{u < v} \mathbf{1}[s_{uv}(m) \neq 0]$. The evaluation includes hydrogen atoms and all atom pairs. For the WBO collection, $\delta_{uv} = 0.3$ for pairs involving a metal and 0.5 otherwise. These thresholds are independent of τ_{iso} .

The decoder returns one mapping witness per distinct signed event class, identifying patterns related by exact source permutations that preserve the event-response labels. First, symmetry-invariance certificates identify families whose mappings all produce equivalent events, and reachable-image bounds exclude families outside a requested event window. Before constraint encoding, adjacent nested action groups are absorbed: if $H \leq G$, then $HG = GH = G$. Exact generator-membership tests remove these redundant choices without changing the final preservation constraints. Remaining families are encoded symbolically in Z3 (3). Each satisfying assignment yields a concrete mapping and its event pattern. A constraint then excludes that entire event-equivalence class, and the process repeats. This projects directly onto event classes rather than explicitly enumerating full mappings or permutation groups.

For a window $E \leq K$, decoding is complete when every retained family is certified exhausted or excluded. A timeout leaves completeness unresolved. This certificate concerns the saved search output within the window; it does not establish exhaustive exploration of all chemically possible mappings. The default deduplication preserves families that contain different event counts, as illustrated in the heavy-atom projection in Figure 1c.

3 Evaluation

3.1 Datasets and configurations

We evaluate annotated mapping recovery on all 1,851 Golden records (6) and event alternatives on 140 native coordinate/WBO endpoint pairs. Golden inputs use formal bond orders and explicit-H expansion. Reference labels are removed before search. Correctness compares the annotated heavy-atom correspondence, including unmatched atoms, modulo endpoint chemical automorphisms preserving charge, isotope, hydrogen count, and stereochemistry. A recovery requires a verified reference witness among the returned alternatives. Unresolved cases remain in the denominator.

The coordinate collection contains balanced endpoint pairs with 18–137 explicit atoms, including organometallic and multicomponent structures. It has no annotated mappings and was inspected during method development. Its evaluation measures coverage of comparator event classes, not chemical accuracy. SLAP outputs are rescored on the same original WBO matrices using Equation (5). Each returned heavy-atom assignment is completed by symbolic hydrogen refinement. The comparator classes are those observed among these returned witnesses; its hydrogen label families are not exhaustively decoded.

Table 1 separates the settings actually evaluated. Golden measures the fragment-search stage; the 140-case evaluation also includes local competition and final event decoding. The competition configuration considers up to eight distinct parent partitions and two successive priority changes. Its budget was selected on development cases before the full 140-case evaluation. Comparator mappings are not used to propose or accept search branches. A two-order run of the final coordinate pipeline was not performed. Detailed protocols and artifact identifiers are given in Appendix A.

3.2 Golden mapping recovery

With one ordering per cut and the single-edge sweep, bidirectional GRAFT recovers 1,834/1,851 annotated mappings (99.08%). Three orderings recover 1,837 (99.24%; Table 2), and ten recover 1,840 (99.41%), with 2

Table 1 Evaluation configurations. Both use $b = 0.2$, $\tau_{\text{iso}} = 1$, explicit H, and uncut plus single-edge searches. The no-sweep controls retain only the uncut graph. A seed setting counts source-atom orderings per cut.

Setting	Golden	Coordinate/WBO
Reactions	1,851	140
Search directions	Both, unioned	Reactant to product
Seed orderings	1, 2, 3, 10	1
Branch cap	100	2,000
Competition completions per reaction	0	Up to 128
Evaluation	Reference-family recovery	Signed event-class coverage

Table 2 Golden recovery with and without the sweep. Recovery uses all 1,851 records. CPU averages use the same 1,821 completed reactions on the Mac, including both directions and the respective sweeps; reference verification is excluded. Dashes mark unmeasured unswept search cost or mixed-host configurations whose times are not compared. SLAP unions binary and weighted modes in both directions; GRAFT uses both directions. “Absent” means excluded from saved families; “unknown” denotes unresolved verification.

Configuration	Recovered	%	Absent	Unknown	CPU s/reaction
GRAFT, 1 seed, no sweep	1,489	80.44	362	0	–
SLAP, no sweep	1,661	89.74	189	1	–
GRAFT, 1 seed + sweep	1,834	99.08	17	0	2.52
GRAFT, 2 seeds + sweep	1,834	99.08	17	0	4.95
GRAFT, 3 seeds + sweep	1,837	99.24	14	0	–
GRAFT, 10 seeds + sweep	1,840	99.41	9	2	–
SLAP sweep	1,796	97.03	52	3	4.82

reactions unresolved. Two orderings also recover 1,834 (99.08%), with no unresolved reactions at either setting. These are family-recovery measurements: they quantify whether a reference correspondence is retained, rather than whether it is ranked first.

Without the sweep, one-seed GRAFT recovers 1,489/1,851 references (80.44%), with no unresolved cases. Uncut SLAP, unioned over both directions and binary/weighted modes, recovers 1,661 (89.74%), with one unresolved case. Adding the sweep recovers 345 additional references for GRAFT and 135 for SLAP. Thus, GRAFT’s higher recovery in the swept comparison depends on the broader search: its one-seed uncut control has lower coverage than the uncut SLAP union. The controls use saved uncut outputs from the same final campaign; no new searches are needed.

The gain is consistent with GRAFT’s greedy fragment growth. Once an extension is accepted, placements incompatible with that larger fragment are discarded; branching over the surviving placements cannot recover them. For example, growing fragment A across an edge (u, v) can exclude a placement that would permit a different neighboring fragment B. Omitting (u, v) in a sweep run removes that adjacency from both growth and fragment-preservation tests, potentially changing where growth stops and which placements survive. The union of these runs therefore explores alternative fragment boundaries while retaining the uncut result. Each run remains greedy, and a single-edge sweep cannot explore every possible partition. Cut-specific seed orderings also contribute trajectory diversity, so the measured gain reflects the complete sweep schedule. Final bond events are always evaluated on the original, uncut endpoint matrices.

The SLAP single-edge sweep recovers 1,796/1,851 references (97.03%) using both directions and binary and weighted modes. One-order GRAFT has a higher verified recovery rate by 2.05 percentage points. The paired comparison contains 43 references verified only by one-order GRAFT and 5 verified only by SLAP. This compares recovery among alternatives under the stated configurations; it is not a comparison of single-answer accuracy or equal-size candidate sets.

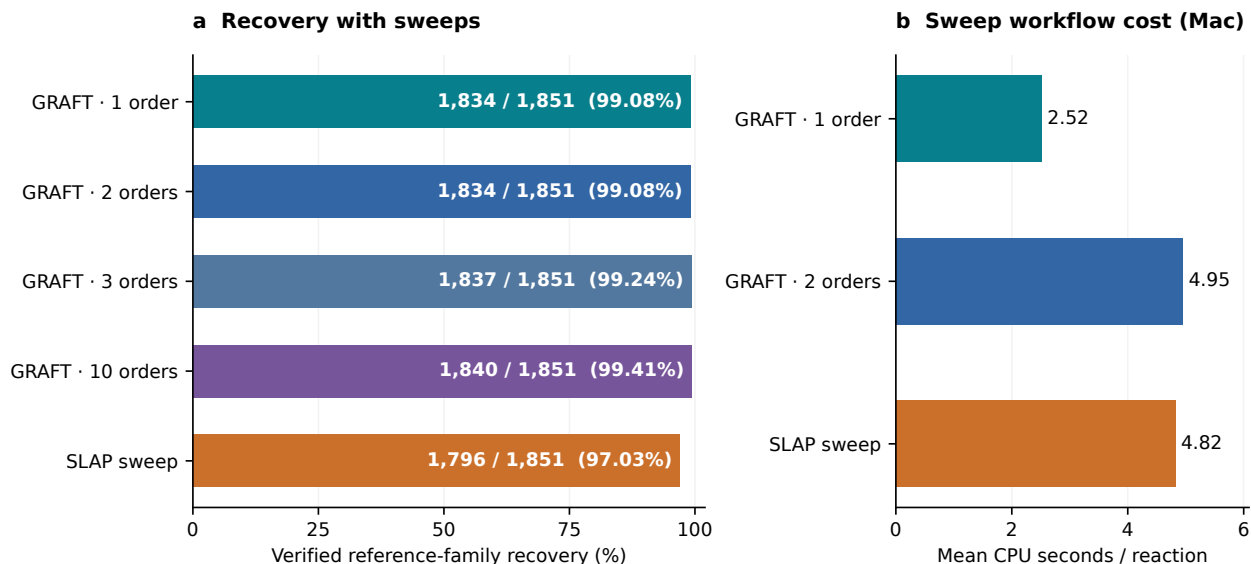


Figure 2 Reference recovery and search cost. (a) Golden family recovery with sweeps for each GRAFT seed setting and SLAP. Unswept controls are reported in Table 2. Every percentage uses all 1,851 records. (b) Mean recorded workflow CPU on the same 1,821 completed reactions on the Mac. Reference verification is excluded; the procedures have different search objectives, output cardinalities, and I/O timing scopes.

On the same-Mac subset of 1,821 reactions with completed one- and two-order searches and complete SLAP output, mean recorded CPU time is 2.52 seconds per reaction for one-order GRAFT, 4.95 for two orders, and 4.82 for the SLAP sweep (Figure 2). Both directions and the respective sweeps are included; reference verification is excluded. GRAFT search-call timing includes checkpoint I/O, whereas SLAP’s graph-construction, mapping, and export timers exclude I/O. These are workflow measurements with different output representations. Three- and ten-order searches span Mac and Linux hosts, so their timings are not pooled into a speedup comparison.

3.3 Released mapping implementations on Golden

We also evaluate default released implementations of RXNMapper (9), LocalMapper (1), Chython (2), Indigo (4), and the Reaction Decoder Tool (RDT) (8), together with default binary and weighted SLAP. Each receives all 1,851 label-free Golden records. Saved predictions are rescored with the same strict heavy-atom relation criterion, retaining unmatched atoms and checking endpoint chemistry. No reference-guided reranking, balancing, or prediction repair is applied.

Table 3 places single-output prediction in context alongside returned-candidate coverage. LocalMapper recovers the strict reference relation in 1,605 reactions, RXNMapper in 1,547, and Chython in 1,561. Default binary SLAP recovers 1,591 among its explicit candidates, compared with 1,796 for the broader bidirectional sweep evaluated above. These default calls use different search budgets from GRAFT; GRAFT’s family recovery is not a top-one accuracy measurement.

RDT recovers 1,030 references under this relation-based criterion, which differs from its chemical-change scoring. Indigo’s 692 recoveries require particular care: 503 outputs contain duplicate heavy-atom map labels, seven alter endpoint chemistry, and 133 calls fail. Chython changes endpoint chemistry in 29 outputs, and three RDT outputs fail validation. These results characterize the pinned implementations and input/output protocol rather than establish universal rankings of the methods. Versions and reproduction details are given in Appendix A.2.

Table 3 Default released implementations on Golden (no sweep). Counts and percentages use all 1,851 reactions, including failures. “First” evaluates the first explicit output; “any” evaluates every explicitly returned candidate. SLAP output order is not asserted to be a top-one ranking, and its compressed label families are not exhaustively decoded here. Invalid counts refer to returned candidates; failed counts refer to reactions with no mapped output after the recorded execution retries.

Implementation	First correct	Any correct	Invalid	Failed
RXNMapper 0.4.3	1,547 (83.58%)	1,547 (83.58%)	0	0
LocalMapper 0.1.5	1,605 (86.71%)	1,605 (86.71%)	0	0
Chython 2.18 / rxnmap 2.0	1,561 (84.33%)	1,561 (84.33%)	29	5
SLAP 1.0.0, binary	1,399 (75.58%)	1,591 (85.95%)	0	0
SLAP 1.0.0, weighted	1,401 (75.69%)	1,564 (84.49%)	0	0
Indigo 1.46.0	692 (37.39%)	692 (37.39%)	510	133
RDT 4.0.0	1,030 (55.65%)	1,030 (55.65%)	3	0

3.4 Bond-event alternatives on coordinate inputs

The configuration with fragment competition returns full explicit-atom mappings for all 140 coordinate pairs. Across the collection, it recovers 166 of the 168 minimum-event classes among the returned SLAP-sweep witnesses, covering every compared class in 139 reactions (Figure 3). In the remaining carbocation reaction, it recovers three of five comparator classes. Both searches reach four events there; the difference concerns two alternative patterns at that minimum. Against native SLAP, coverage is 155/160 minimum-event classes, again with all missing classes confined to the same reaction.

Local competition expands the retained event-class set by 36 classes across 9 reactions within the evaluated windows. One added event class in one reaction requires decoding internal family shuffles and is absent from the repair representatives. This demonstrates why final candidate extraction must retain correlated alternatives rather than score only one witness. All accepted competition families were completely decoded within their corresponding event windows; this does not imply that every search trajectory was explored.

3.5 Final representation and postprocessing cost

Unordered fragment-pair deduplication reduces the 140-case total from 237,645 ordered branch records to 124,641 final branches, a 47.55% reduction. The median falls from 596.5 to 397 branches per reaction. The largest individual reduction is from 52,669 to 2,856 (Figure 3).

The final representation retains 236,653 distinct normalized mapping-family records. Thus, fewer branch identities do not imply an equal reduction in event-decoding work: several constrained families can belong to the same fragment combination. Catalogue construction, shared intrinsic symmetry computation, representative checks, and complete event-window decoding together take 11.60 minutes of wall time and 25.70 recorded CPU-minutes on three local workers. Peak combined sampled memory is 1.88 GiB. These measurements include certificate journaling and bounded continuation, and exclude matching and competition. Other benchmark jobs shared the computer, so the wall time is an observed campaign duration rather than an isolated latency measurement.

Every family in all 140 catalogues was freshly decoded or certified within its specified event window (Appendix A). The output contains 336 distinct reaction-specific event classes, each with a mapping witness. No case remains unresolved within those windows. Preservation of the family union follows from the transformation in Equations (3) and (4).

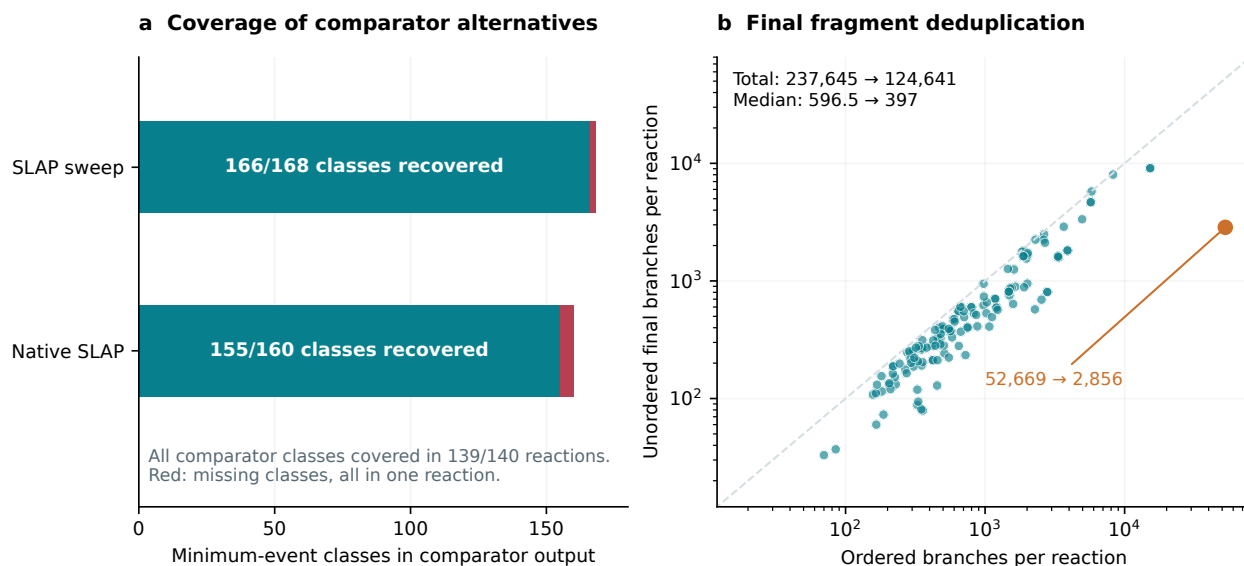


Figure 3 Event alternatives and final branch representation on 140 reactions. (a) Recovery of minimum-event classes among fresh comparator witnesses. GRAFT covers every SLAP-sweep class in 139 reactions; the remaining reaction has two missing alternatives at the same four-event minimum. (b) Per-reaction ordered and unordered branch counts; the diagonal denotes equality. The highlighted case falls from 52,669 to 2,856. Branches are fragment-pair combinations, not counts of explicit mappings or event classes.

4 Discussion

GRAFT separates structural search, symmetry representation, and interpretation of bond changes. Continuous growth accumulates context while preserving multiple placements; local competition allows neighboring fragments to challenge an established partition. Final grouping removes irrelevant growth order, while the flat family representation retains constraints and correlated permutations that can affect the output. Symbolic projection then exposes distinct bond-event hypotheses without expanding every mapping.

The Golden results support broad reference coverage by the fragment-search stage, and the coordinate results show that the competition configuration retains nearly all minimum-event alternatives found by the SLAP sweep. These are complementary measurements. The coordinate collection has no mapping annotations, and event minimization is not an energy model or a proof of mechanistic plausibility. The priority budget was tuned on development cases, so independent validation is needed to establish generalization.

Finite schedules, greedy growth, branch caps, and competition budgets limit coverage. A complete event-window certificate applies only to the saved families. Matching-preserving automorphisms can change finer-threshold events. GRAFT therefore returns inspectable alternatives, with explicit completeness scope, for chemical review and subsequent calculations.

Code and data

Source code, benchmark summaries, and figure-generation scripts accompany the repository at https://github.com/yunhzou/coordinate_alignment, branch `paper/continuous-fragment-growth`. Golden is distributed by its original authors (6). Per-case results and the configurations used for each measurement are supplied with this manuscript.

References

- [1] Shuan Chen, Sunggi An, Ramil Babazade, and Yousung Jung. Precise atom-to-atom mapping for organic reactions via human-in-the-loop machine learning. *Nature Communications*, 15:2250, 2024. doi: 10.1038/s41467-024-46364-y.
- [2] Chython contributors. Chython: molecular and reaction processing toolkit. <https://github.com/chython/chython>. Evaluated version 2.18 with chython-rxnmap 2.0; model hash archived.
- [3] Leonardo de Moura and Nikolaj Bjørner. Z3: An efficient SMT solver. In *Tools and Algorithms for the Construction and Analysis of Systems*, pages 337–340. Springer, 2008. doi: 10.1007/978-3-540-78800-3_24.
- [4] EPAM Systems. Indigo cheminformatics toolkit. <https://github.com/epam/Indigo>. Evaluated version 1.46.0.
- [5] Shin-ichi Koda and Shinji Saito. General and scalable atom-to-atom mapping via Weisfeiler–Lehman-like approximate graph matching. *ChemRxiv*, 2025. doi: 10.26434/chemrxiv-2025-hthwn. Preprint, version 1, 25 November 2025.
- [6] Arkadii Lin, Natalia Dyubankova, Timur I. Madzhidov, Ramil I. Nugmanov, Jonas Verhoeven, Timur R. Gimadiev, Valentina A. Afonina, Zarina Ibragimova, Assima Rakhimbekova, Pavel Sidorov, Andrei Gedich, Rail Suleymanov, Ravil Mukhametgaleev, Joerg Wegner, Hugo Ceulemans, and Alexandre Varnek. Atom-to-atom mapping: A benchmarking study of popular mapping algorithms and consensus strategies. *Molecular Informatics*, 41(4):2100138, 2022. doi: 10.1002/minf.202100138.
- [7] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60: 94–112, 2014. doi: 10.1016/j.jsc.2013.09.003.
- [8] S. A. Rahman, G. Torrance, L. Baldacci, S. Martínez Cuesta, F. Fenninger, N. Gopal, S. Choudhary, J. W. May, G. L. Holliday, C. Steinbeck, and J. M. Thornton. Reaction decoder tool (RDT): extracting features from chemical reactions. *Bioinformatics*, 32(13):2065–2066, 2016. doi: 10.1093/bioinformatics/btw096.
- [9] Philippe Schwaller, Benjamin Hoover, Jean-Louis Reymond, Hendrik Strobelt, and Teodoro Laino. Extraction of organic chemistry grammar from unsupervised learning of chemical reactions. *Science Advances*, 7(15):eabe4166, 2021. doi: 10.1126/sciadv.abe4166.

A Reproducibility details

A.1 Search and comparison protocols

The active-edge floor is $b = 0.2$ and the matching tolerance is $\tau_{\text{iso}} = 1.0$. Seed streams are deterministic with root seed 42. Sweeps retain the uncut search and individual source-edge deletions. Bidirectional Golden output is a union of returned relations; branches from opposite directions are not spliced. Growth and branch caps bound compressed candidates rather than explicit mapping counts.

Golden uses dataset revision 793475e. Its heavy-atom evaluator compares correspondence relations modulo endpoint chemical symmetry, retains unmatched atoms, and rejects altered endpoint chemistry. Hydrogen atoms participate in search but their individual identities are not reference labels. All 1,851 records remain in the denominator. All reported GRAFT configurations were rerun against the final algorithm source, commit 3a9ef9a. Execution manifests and component hashes accompany the per-case results. Golden evaluates the fragment-search stage, while the coordinate protocol also evaluates competition and event decoding.

The no-sweep GRAFT control retains only the original uncut checkpoint at one seed in each direction. Its 3,702 saved graphs are finalized and evaluated using the same reference-family criterion, yielding 1,489 recoveries and 362 completed non-recoveries. For SLAP, the control retains ordinal-zero calls from both directions and both bond modes; incomplete sweep records are checked at that ordinal separately. Case 1723 has no committed reverse uncut output and remains unresolved. These are paired controls of the final campaign, not substituted historical configurations. Re-evaluation time is excluded from search-cost claims.

Each search or evaluation subprocess has a 300-second watchdog and an initial 3 GiB sampled-memory limit. Externally interrupted phases can resume their completed cuts for at most two bounded continuations with a 6 GiB allowance. Golden reference evaluation has a 220-second internal budget. Completed raw cuts can also be finalized and verified individually, without new search. A positive result requires a verified witness; it can establish recovery even when another direction or the combined archive is incomplete. A negative result requires completed verification of the saved output; other cases remain unknown.

The campaign begins on an Apple M4 Pro Mac with 14 logical CPUs and 48 GiB RAM, using at most eight single-core workers, and continues on Linux CPU nodes with a final concurrency limit of 150 single-thread workers. Python 3.12.14, RDKit 2026.3.6, pynauty 2.8.8.1, SymPy 1.14.0, and Z3 5.1.0.0 are fixed across hosts. Native extensions are rebuilt on Linux; source-hash checks, focused tests, and two search/reference parity cases verify the port. Numerical-library threads are fixed to one. CPU averages use 1,821 common reactions with both one- and two-order searches completed on the Mac and complete comparator output. Search-call timing includes checkpoint I/O and excludes imports, preparation, reference evaluation, and interrupted work; SLAP’s instrumented workflow excludes I/O. Higher-order searches use mixed hosts and are excluded from this timing comparison.

The SLAP comparator uses SLAPMapper 1.0.0, revision ea248fd, with both directions and binary and weighted modes. Each sweep variant uses one original ordering and rebuilds initialization for the uncut input and each single-edge deletion. Returned labels are restored to the original endpoint structures before evaluation. Its 1,796 verified Golden recoveries leave 52 completed misses and 3 unresolved records; all remain in the denominator. The sweep perturbs SLAP’s input graph and assignment objective, whereas GRAFT relaxes fragment-preservation constraints.

On the coordinate collection, the exact original XYZ-derived SLAP adjacency matrices are restored from the saved graph records and verified against every source-edge deletion. Atom order and atomic numbers are retained; mapping labels are reset. Fresh native and bidirectional single-edge SLAP runs use binary graphs and the original initialization. The XYZ-to-graph preparation itself is not rerun. GRAFT uses the cached WBO matrices. Each returned SLAP heavy-atom assignment receives a symbolic hydrogen refinement under the comparator’s original bond-change objective, and its witness is rescored on the original WBO matrices using Equation (5). Tied hydrogen optima can give different signed-event classes; the reported comparator set comprises the returned witnesses, not the entire hydrogen label relation. This compares the stated endpoint workflows rather than isolated solvers on identical graph representations.

A.2 Default comparator reproduction

The default comparator predictions were generated with RXNMapper 0.4.3, LocalMapper 0.1.5, Chython 2.18 with chython-rxnmap 2.0, Indigo 1.46.0, SLAPMapper 1.0.0 at revision [ea248fd](#), and RDT 4.0.0 at revision [0d663225](#). Chython’s released ONNX model is identified by its saved hash; it is not represented as a reproduction of a historical GraphormerMapper checkpoint. RXNMapper and LocalMapper use their released default models on CPU. SLAP uses native output order with its default heavy-atom symmetry handling, with binary and weighted settings reported separately. Indigo uses its native automapping interface; RDT uses the public `RDT.map` API.

The complete archive retains mapping strings, explicit candidate order, version/model hashes, and per-record evaluations. All 12,957 method/reaction records were rescored with the current evaluator and agree with the archived outcomes. Dataset and label-free input hashes match the final GRAFT campaign. Agent-field components returned by a mapper are retained on the input side, preserving their original atom labels. Changed endpoint chemistry and duplicate heavy-atom labels invalidate a prediction. All 1,851 records remain in each denominator.

RDT’s final records include execution retries after a worker failure; every final call returns an output. Invalid outputs are not repaired. These default implementations were timed on the original cluster, so their archived call times are not included in the same-Mac comparison in [Figure 2](#). Full timing and failure records accompany the reproducibility archive.

A.3 Bounded fragment competition

The coordinate configuration uses one seed ordering, cap 2,000, up to eight lowest-event distinct parent source partitions, at most two successive priority changes, and at most 128 proposal-growth calls and 128 completion calls per reaction. Smaller challenger fragments are visited first, with deterministic contact order. A proposed challenger is rematched with a single contacting source atom. Conflicting source assignments and occupied target images are released; attached components of at most eight atoms can also be released around the affected region. Remaining assignments are anchors for completion.

The admission check evaluates every complete representative against the original endpoint weights and its required fragment constraints. A completion graph is excluded if any checked complete representative violates these constraints. Only accepted graphs contribute to reported coverage. The ordinary search output is retained in union. Pending queue entries, omitted parent partitions, and capped continuations are not covered by decoding certificates.

Competition and decoding use five-minute per-case watchdogs with persisted results. Complete family certificates survive continuation, and unfinished families remain explicit. The full coordinate search, competition, and decoder stages are rerun from the original endpoint inputs. Timing for complete final decoding includes all of its bounded passes; it excludes the preceding search and competition stages.

A.4 Event windows and output equivalence

A signed event class contains the nonzero atom-pair signs from [Equation \(5\)](#), modulo source permutations preserving the full event-response coloring relative to the opposite endpoint. This coloring retains every threshold comparison needed to transport a pattern without changing its identity. It is separate from the intrinsic fragment matching symmetry in [Equation \(2\)](#). The exported witness records both the atom mapping and the formed/strengthened and broken/weakened pair lists.

The 140-case comparison uses fixed per-case windows from the evaluation protocol. Each limit is one event above the larger of the native-SLAP and SLAP-sweep minima under the common signed-event definition. All 236,653 retained flat families are exhausted in these windows without a class-count cap; higher-event patterns are outside the completeness claim.

Table A1 Complete final decoding on all 140 reactions. Every retained family is certified within the listed per-case event limit.

Maximum events	1	2	3	4	5	6	7	8	9
Reactions (all complete)	6	11	39	32	33	10	7	1	1

For the saponification example in Figure 1, the direct assignment $(1, 2, 3, 4, 5, 6) \mapsto (a, b, c, d, e, f)$ loses O4–C5 and gains C5–O6. Exchanging the images of O3 and O4 additionally weakens C2–O3 and strengthens C2–O4: two versus four heavy-atom events. Both assignments preserve the internal matching constraints of fragment $\{1, 2, 3, 4\}$ at $\tau_{\text{iso}} = 1$. The challenger instead assigns O4 to f and O6 to d, retaining O4–C5 and replacing C2–O4 by C2–O6. Formal bond orders define this illustration; the displayed event counts include only heavy atoms. Both product acetate oxygens have zero attached hydrogens. The figure builder verifies these mappings and event lists. For the separate alcohol-site tree, it verifies every attached hydrogen and every bond inside each local matched fragment, the distinct target-site environments, and injectivity along all six displayed paths.

A.5 Lossless flattening

For each complete search path, flattening retains its terminal representative, the union of required fragment-bond constraints, and the allowed action sequence with prefix locks already applied. It removes only representation choices that leave the admissible relation unchanged. Reordering disjoint-support factors preserves their product; adjacent nested factors satisfy $HG = GH = G$ when $H \leq G$. Subgroup containment is checked by exact generator membership using a stabilizer chain. Other factors with overlapping support retain their order. Identical resulting records are shared, and final branch grouping stores the union of their family identifiers. Consequently, the union of mappings before and after grouping is the same. The procedure does not infer equality merely from a shared fragment partition or a shared automorphism group.

The event decoder obtains completeness by invariance certificates, valid lower bounds, or an unsatisfiability result after excluding every discovered output class. Budget exhaustion is recorded as incomplete. This distinguishes preservation of a saved mapping union, exhaustive decoding of a specified event window, and exhaustive atom-mapping search.